Xi'an Jiaotong-Liverpool University
西交利物浦大學

Module Code: CHE105
Module Title: Introductory Organic Chemistry for Non-Chemists
Semester 1, 2023/24 Year 2

FINAL EXAMINATIONS

Duration: 2 hours

Module Examiner: Dr. E.Amigues **Department: Chem. Tel: 0512 8816 7123**

Instructions to Candidates:

1. Answer all questions.

2. The maximum mark is 100.

3. You can answer the questions in any order you prefer, e.g. start from Q4.

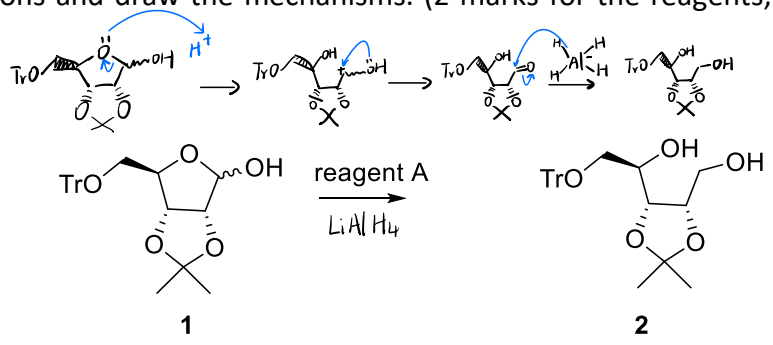
4. Answers should be written on the Answer Booklet provided; the left-hand page of the Answer Booklet can be used for drafting work when necessary.

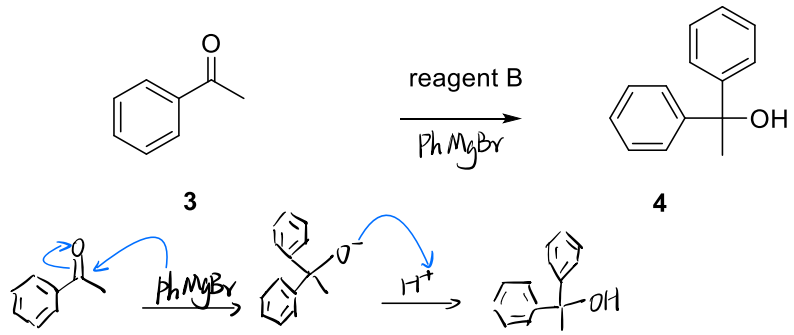
5. Answers should be written using pen, black or blue, **NO PENCIL**.

6. Only solutions written in English are accepted.

Question 1 (10 marks)

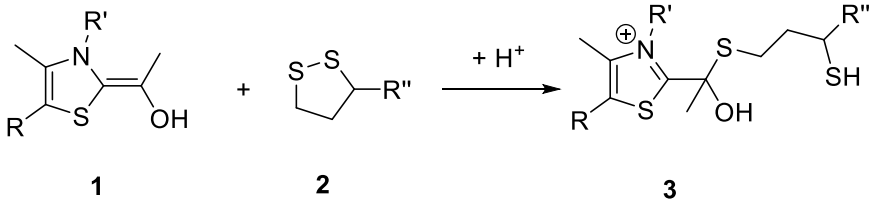
Identify the suitable reagents you should use in order to perform the following transformations and draw the mechanisms. (2 marks for the reagents, 3 marks for the mechanism)

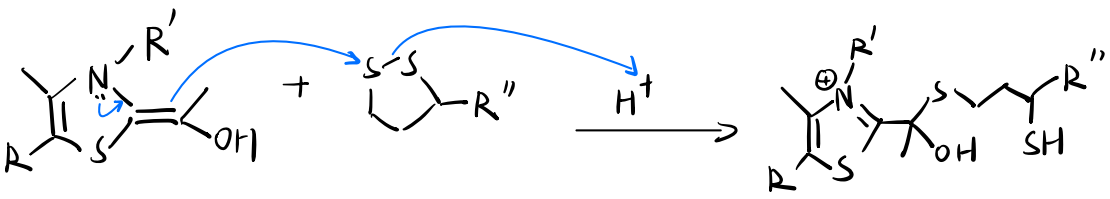




Question 2 (10 marks)

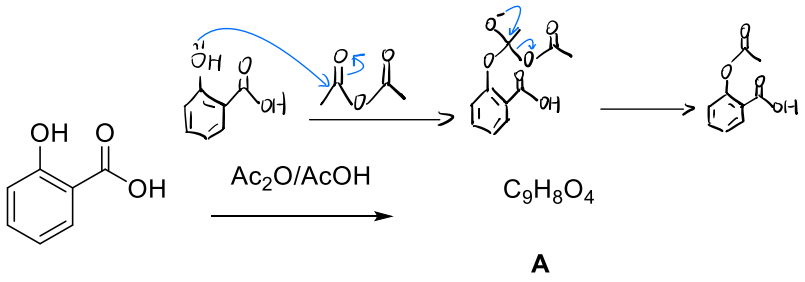
The following reaction is part of the sequence by which pyruvate is converted to acetyl CoA. Propose a mechanism

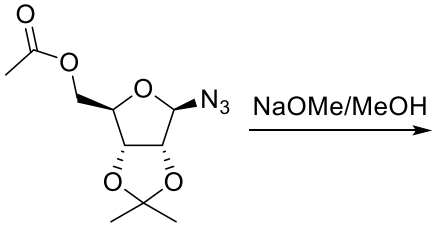


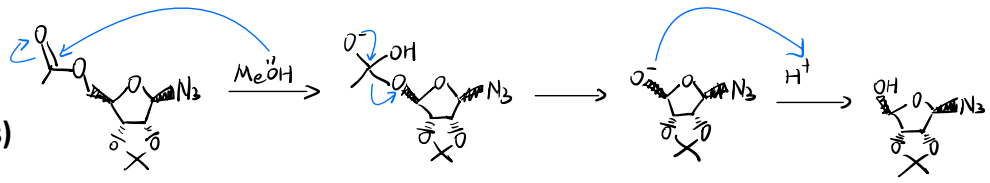


Question 3 (10 marks)

Identify the products **A** and **B** that will be formed in reactions below and draw the mechanisms. (5 marks for each reaction, 2 marks for the product, 3 marks for the mechanism)

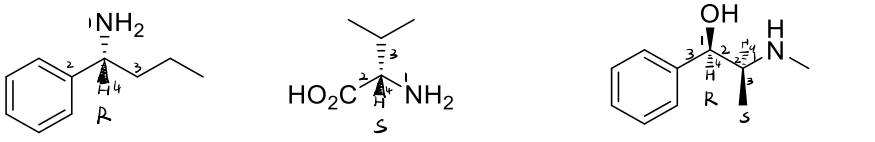






Question 4 (20 marks)

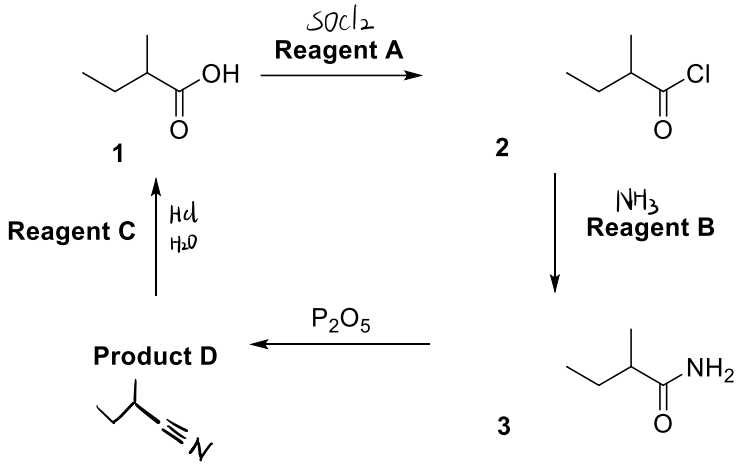
Assign the configuration (*R* or *S*) to each of the stereogenic centres in the three molecules below. Clearly show how you have worked out your answer (10 marks for the assignment, 10 marks for the justification)



Show the CIP priority rules for every single stereocenter

Question 5 (10 marks)

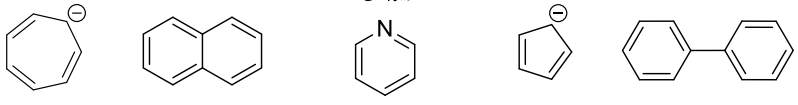
Identify the reagents **A, B, C** and the product **D**. No mechanisms required. (2.5 marks each)



Question 6 (10 marks)

Which structural features does a compound need to have in order to be aromatic? (5 marks). Which of these compounds are aromatic? Justify your answer with electron counting. (5 structures, 1 marks each)

① cyclic structure
② Planarity
③ conjugation (π electrons must be delocalized over the entire ring)
④ Follow Hückel's rule (4n+2π electrons)

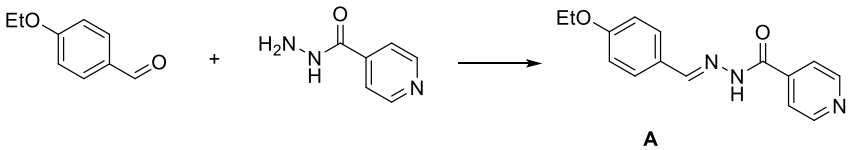


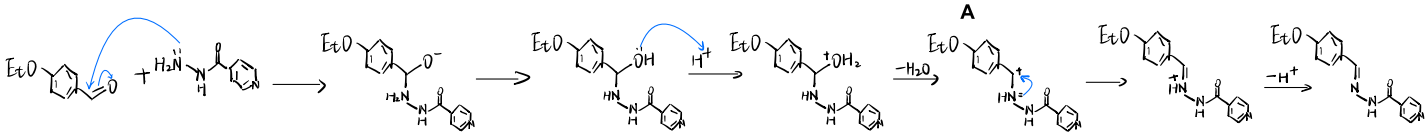
1 2 3 4 5

1、 8π electrons, not aromatic
2、 10π electrons, planar, every atom involved
3、 6π electrons, planar, every atom involved
4、 6π electrons, planar, every atom involved
5、 not cyclic structure

Question 7 (20 marks)

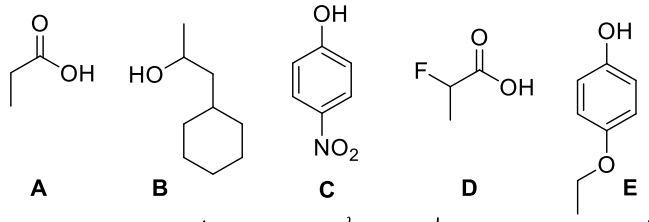
Suggest mechanisms for the reactions that lead to the formation of product **A**. (20 marks)





Question 8 (10 marks)

Rank compounds **A** to **E** in order of increasing acidity. Explain your answers.



A **B** **C** **D** **E**

B < E < C < A < D

reason = ① acidity : carboxylic acids > Phenol > Alcohol ∴ A > E > B ∴ D > A

② Inductive effect = F (fluorine) withdraw electron density, stabilizing the conjugate base

③ Resonance effect : NO_2 can delocalize negative charge through resonance increasing acidity ∴ C > E

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